

Élise Kerdoncuff

Education and Professional Experience

- 2025-current **Postdoc**, Human Evolutionary Genetics team at Pasteur Institute, Paris, France. Tracing Ancestral Genomes from Genome-Wide Genealogies: Insights Into Human Genetic History in the Pacific.
- 2022-2025 **Postdoc**, Moorjani Lab at University of California, Berkeley, USA. Study of South Asian population evolutionary history and development of local ancestry inference methods.
- 2017-2021 **PhD**, *Detection of Recent Decline in Population by a Genomic Approach*, Stochastic Models for the Inference of Life Evolution group (CIRB, Collège de France) and Atelier de Bio-Informatique group (Muséum National d'Histoire Naturelle - MNHN). Supervised by Guillaume Achaz and Amaury Lambert, Sorbonne University.
- 2015-2017 **Master**, Ecology, Biodiversity, Evolution, speciality: Evolutionary Ecology, Sorbonne University.
- 2012-2015 **Bachelor**, Bi-disciplinary bachelor of biology and mathematics, Sorbonne University at the Roscoff Marine Station, France.

Publications, Preprints

1. **Kerdoncuff, E.***, Marohn, M.*, Cramer, N., Dey, S., Kardia, S., Thangaraj, K., Ségurel, L., Lee, J., Dey, AB., Moorjani, P. Revisiting the Evolution of Lactase Persistence: Insights from South Asian Genomes *bioRxiv preprint* (*EK and MM contributed equally to this work)
2. Tian, D., **Kerdoncuff, E.**, Wilson, K., Feuerbacher, O., Schwemm, M.R., Gumm, J., Moorjani, P., Martin, C.H. Estimate of the mutation rate in the endangered Devils Hole pupfish provides support for the drift-barrier hypothesis at an outlying extreme. *bioRxiv preprint*
3. **Kerdoncuff, E.***, Skov, L.*, Patterson, N., Zhao, W., Lueng, Y.K., Schellenberg, G.D., Smith, J., Ganna, A., Dey, AB., Kardia, S., Lee J., Moorjani, P. 50,000 years of Evolutionary History of India: Insights from 2,700 Whole Genome Sequences. *Cell* (*EK and LS contributed equally to this work)
4. **Kerdoncuff, E.**, Ianni-Ravn, M., Roze, D., Novembre, J., Lambert, A., Achaz, G. How demography shapes linkage disequilibrium with or without recombination. *bioRxiv preprint*
5. Freund, F.*, **Kerdoncuff, E.***, Matuszewski, S., Lapierre, M., Hildebrandt, M., Jensen, J. D., Ferretti, L., Lambert, A., Sackton, T. B., Achaz, G. Interpreting the pervasive observation of U-shaped Site Frequency Spectra. (2023) *PLOS Genetics* (*FF and EK contributed equally to this work)
6. Foutel-Rodier, F., Blanquart, F., Courau, P., Czuppon, P., Duchamps, J., Gamblin, J., **Kerdoncuff, E.**, Kulathinal, R., Régnier, L., Vuduc, L., Lambert, A., Schertzer, E. From individual-based epidemic model to McKendrick-von Foerster PDEs: a guide to modeling and inferring COVID-19 dynamics. (2022) *Journal of Mathematical Biology*
7. **Kerdoncuff, E.**, Lambert, A., Achaz, G. Testing for population decline using maximal linkage disequilibrium blocks. (2020) *Theoretical Population Biology*.
8. Ribardièrre, A., Centanni, J., Dano, A., Coudret, J., Daguin-Thiébaud, C. Houbin, C., **Kerdoncuff, E.**, Jambut, S., Cordaux, R., Broquet, T. Female-biased sex ratios unrelated to *Wolbachia* infection in European species of the *Jaera albifrons* complex (marine isopods). (2018) *Journal of Experimental Marine Biology and Ecology*.

Presentations

Invited

- *Gokcumen Lab Seminar*, October 2025, Online seminar, University of Buffalo, New-York, USA.
- *Anthropologie Génétique UMR seminar*, March 2025, Musée de l'Homme, Paris, France.
- *Human Evolutionary Genetics group weekly seminar*, May 2024, Pasteur Institute, Paris, France.
- *Workshop on Li-Stephens based methods and applications*, December 2022, Harvard, USA.
- *Center for Theoretical and Evolutionary Genetics weekly seminar*, April 2022, UC Berkeley, USA.
- *Workshop : Evolutionary Genomics and Life History Traits*, October 2019, Montpellier, France.
- *Séminaire du CESCO*, June 2019, MNHN, Paris, France.

Contributed talks

- *Probabilistic Modeling in Genomics*, March 2025, Cold Spring Harbor, USA.
- *Bay Area Population Genomics*, November 2024, Berkeley, USA.
- *New Frontiers in Reconstructing Human Evolution History*, June 2023, Hannover, Germany.
- *Bay Area Population Genomics*, October 2022, Berkeley, USA.
- *Popgroup53*, January 2020, Leicester, Royaume-Uni.
- *GDR Approches Interdisciplinaires en Evolution Moléculaire (AIEM)*, November 2019, Toulouse, France.
- *Annual Meeting Society for Molecular Biology and Evolution (SMBE), Symposium: The Coalescent in the Era of Population-Scale Genomics*, July 2019, Manchester, Royaume-Uni.
- *Mathematical and Computational Evolutionary Biology (MCEB)*, May 2019, Porquerolles, France.
- *Séminaire de l'ISYEB*, May 2019, MNHN, Paris, France.
- *CIRB seminar*, February 2019, Collège de France, Paris, France.
- *Popgroup52*, January 2019, Oxford, Royaume-Uni.

Poster

- *Annual Meeting Society for Molecular Biology and Evolution (SMBE)*, July 2024, Puerto Vallarta, Mexico.
- *Population, Evolutionary, and Quantitative Genetics Conference*, June 2022, Asilomar, USA.
- *Probabilistic Modeling in Genomics*, October 2019, Aussois, France.
- *GDR AIEM*, November 2018, Lille, France.
- *MCEB*, June 2018, Saint-Martin-de-Londres, France.
- *GDR AIEM*, November 2017, Lyon, France.

Media

2024-2025 Interviews on the Indian Evolutionary History:

- General audience newspapers: *Times of India*, *San Francisco Chronicle*
 - Scientific newspapers: *New scientist*, *Live Science*, *QB3 Berkeley News*
 - Radio/Podcast : *Sher-E-Punjab AM 600 Vancouver*, *The Coffee House Experience Podcast*
- 2021 Podcast episode of *Chercheuses* on my PhD journey: [In French](#)

Teaching Experience

- 2025 *Université Paris Cité*, Population Genetics, Teaching Assistant (20h).
- 2022 *University of California, Berkeley*, Quantitative Biology Bootcamp, Teaching Assistant (15h).
- 2022 *University of California, Berkeley*, Computational Biology Skill Seminar, lecture (1h, [Video](#)).
- 2018-2019 *Sorbonne Université, Paris*, Python, Teaching Assistant (40h).
- 2017-2018 *IUT de Vélizy*, C Language, class and tutorials (45h).

Mentoring/Supervision

- 2026 **Zélie Solan De Renty**, M2 EVOGEM Sorbonne University, 6 months.
Imputation of ancient DNA from Eastern Polynesia.
- 2023-2025 **Meaghan Marohn**, PhD Candidate UC Berkeley.
Revisiting the evolution of lactase persistence : insights from South Asian genomes.

- 2024 **Mara Baylis**, Rotation student UC Berkeley, 3 months.
Description, origin and functional impact of the haplotype carrying the BTNL2 gene inherited from Denisovan and present in human populations.
- 2021 **Pierre Imbert**, M2 Bio-Informatique Paris Diderot, 6 months.
Benchmark of inference methods: a focus on population decline.
- 2019 **Ludovic Fourteau**, M1 Bio-Informatique Paris Diderot, 3 months.
Effect of a single recombination event on the diversity in a population.
- 2019 **Rebecca Clodion**, M1 Bio-Informatique Paris Diderot, 3 months.
The mysterious U spectrum: a population analysis of the genetic diversity.

Grants/Scholarship

- 2024 Symposium SMBE 2024 (Organisation grant 4,250\$).
- 2023 Keystone Symposia Future of Science Fund scholarship (Travel grant 1,200\$).
- 2020-2021 ATER Collège de France grant for a fourth year of PhD (Salary >20,000€).
- 2017-2020 Interfaces pour le vivant (IPV) doctoral school grant for a three year PhD (Salary >50,000€).

Service

Reviewer: *Genome Biology and Evolution* (2026), *PNAS* (2026), *Philosophical Transactions B* (2026), *Nature communications* (2025), *Human Genetics and Genomics Advances* (2025), *BMC Biology* (2025), *American Journal of Human Genetics* (2024), *Molecular Biology and Evolution* (2024), *Human Population Genetics and Genomics* (2024-2025), *European Journal of Human Genetics* (2022-2023)

Symposium organiser: *Haplotype-based methods and frameworks for inference of evolutionary history*. SMBE 2024, Puerto Vallarta, Mexico.

Student/postdoc representative of the ISYEB (elected member): Institute meetings participation. Organisation of monthly seminars and yearly Young Research day; from 2018 to 2021.

Jury: Best poster and best presentation of the symposium 'Evolutionary Ecology' of the 6th Young Natural History Scientists' Meeting, 12-16 mars 2019, MNHN, Paris, France.

Volunteer at conferences: Evolution 2018, Montpellier, France.

Student representative of the SDUEE Master (elected member): Council meetings participation; from 2016 to 2018.

Computer skills

Languages C, PYTHON

Familiar with the usage of a computer cluster

Repositories

- Scripts of paper *50,000 years of Evolutionary History of India* (github.com/MoorjaniLab/LASI-DAD)
- Scripts and data of paper *Interpreting the pervasive observation of U-shaped Site Frequency Spectra* (github.com/fabfreund/usfs_mmc)
- Code of the four gamete test chopping algorithm into Maximum Lindage Disequilibrium blocs (github.com/Ekerdoncuff/algo_4GT_genome)

References/Collaborators

- Lluís Quintana-Murci, Postdoc advisor, Professor at *College de France* & Research Director at *Institut Pasteur*: lluis.quintana-murci@pasteur.fr
- Etienne Patin, Postdoc advisor, Research Director at *Institut Pasteur*: etienne.patin@pasteur.fr
- Priya Moorjani, Postdoc advisor, Assistant Professor at *UC Berkeley*: moorjani@berkeley.edu
- Guillaume Achaz, PhD advisor, Professor at *Université Paris-Cité*: guillaume.achaz@mnhn.fr
- Amaury Lambert, PhD advisor, Professor at *École Normale Supérieure*: amaury.lambert@ens.fr